

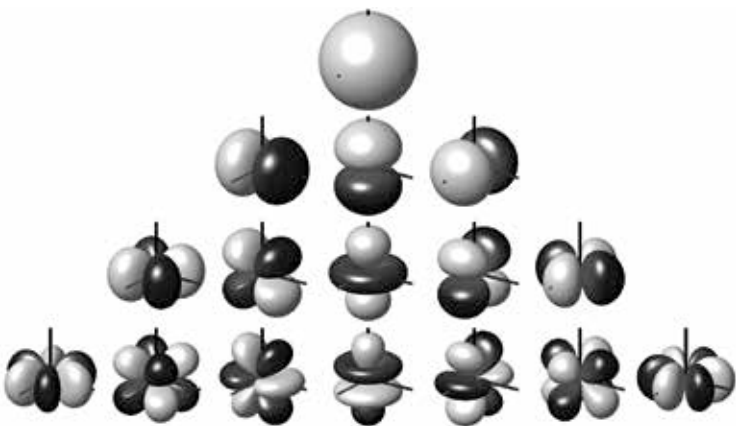
Any topic about a game engine would not be complete without the Unreal Engine. The powerhouse that Unreal Engine 4 is, it lends its full capabilities to VR and completely supports development for the Oculus Rift, GearVR, Playstation VR and other platforms. Infact, they have a highly detailed documentation and training structure available at <http://dgit.in/UnrealVR>. The power of the Unreal Engine community combined with Unreal Engine's expertise with AAA games and even film development makes this one a go to choice for VR development, if you are ok with Unreal's 5% Royalty rule.

Note: If these engines don't make the cut for you, there are a number of other options like Torque3D, Coppercube, FabricEngine etc. It is always better to explore all the alternatives before you begin development.

Audio for VR

Sounds play a crucial role in completing the basic VR experience loop. We have explored Binaural Audio in previous chapters, but simple Binaural audio just doesn't cut it for a complete VR experience. For starters, most binaural audio is recorded statically, as in, it doesn't account for the listeners moving their heads, which can disrupt the immersion badly. This is where we bring in Ambisonics.

Ambisonics is a full surround sound technique which captures the sound on the horizontal plane as well as the sound from above and below the listener. The unique thing about ambisonic sound files is that it's transmission



This image shows the speaker components upto 3rd order (Six directions of coverage) for the B-Format sound used in ambisonics